(Year 2 / A2 Level)

Temperature Lecture Notes

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Chapter 14
Temperature

14.1 Thermal physics basics

14.1.1 Temperature scales

- Celsius scale (unit: °C)

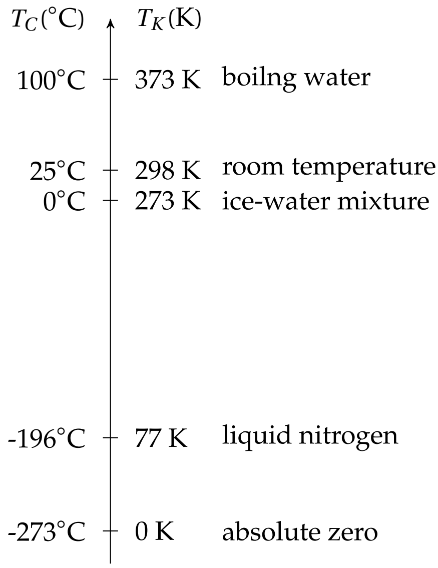
  0°C defined as temperature of ice-water mixture

  100°C defined as temperature of boiling water

- Kelvin scale (unit: K)

  0 K (*absolute zero*) is lowest temperature possible

- conversion rule: $T_K(\text{K}) \underset{+273}{\overset{-273}{\rightleftharpoons}} T_C(^\circ\text{C})$
- change of 1°C equals change of 1 K



14.1.2 Kinetic theory of matter

there are three common states of matter: solid, liquid and gas

they have very different physical properties (density, compressibility, fluidity, etc.)

but deep down, they are all composed of a large number of small molecules

in the **kinetic theory of matter**, we look at microscopic behaviour at molecular level (arrangement, motion, intermolecular forces, separation, etc.)

microscopic behaviour of molecules cause differences in *macroscopic* properties of matter

- solid: molecules close together, tightly bonded, vibrate about their positions
- liquid: molecules quite close together, vibrate but has some freedom to move about
- gas: molecules widely separated, free from neighbours, move rapidly

14.1.3 Specific latent heat

it requires heat energy to *melt* a solid or *boil* a liquid

melting and boiling usually occur at a fixed temperature

thermal energy to cause the change of state at a constant temperature is called *latent heat*

amount of latent heat needed depends on mass of substance: $\boxed{Q = Lm}$

we define **specific latent heat** (L) as the thermal energy required to change the state of *unit* mass of substance with no change in temperature is called

- unit of specific latent heat: $[L] = \mathrm{J \cdot kg^{-1}}$
- specific latent heat is an *intensive* property

i.e., L does not depend on size or shape of sample, L depends on type of substance only ➢

for melting, L is called *specific latent heat of fusion*

for boiling, L is called *specific latent heat of vaporisation*

- latent heat is related to breaking bonds and increasing intermolecular separation

vaporisation requires larger increase in particle separation than fusion

for a given substance, $L_{\text{vapour}} > L_{\text{fuse}}$

Example 14.1 A 3.0 kW electric kettle contains 0.5 kg of water already at its boiling point. Neglecting heat losses, determine how long it takes to boil dry. ($L_{\text{water}} = 2.26 \times 10^6$ J kg^{-1})

✍ heat required: $Q = mL = 0.50 \times 2.26 \times 10^6 = 1.13 \times 10^6$ J

time needed: $t = \dfrac{Q}{P} = \dfrac{1.13 \times 10^6}{3.0 \times 10^3} \approx 380\text{ s} \approx 6.3\text{ min}$ □

Example 14.2 A student measures specific latent heat of fusion for ice. He uses an electric heater to melt ice but the insulation is not perfect. The experiment is carried out twice, with the heater operating at different powers. Use the data table to calculate specific latent heat of fusion.

	Power (W)	time interval (min)	mass of ice melted (g)
test 1	60	3.0	40.4
test 2	90	3.0	56.6

✍ there exists heat gain from surroundings, so effective power $P_{\text{eff}} = P_{\text{heater}} + P_{\text{sur}}$

heat energy to melt ice: $Q = mL = (P_{\text{heater}} + P_{\text{sur}})t$

$$\begin{cases} 40.4 \times L = (60 + P_{\text{sur}}) \times 3.0 \times 60 \\ 56.6 \times L = (90 + P_{\text{sur}}) \times 3.0 \times 60 \end{cases} \Rightarrow \begin{cases} L \approx 333\ \text{J g}^{-1} \\ P_{\text{sur}} \approx 14.8\ \text{W} \end{cases}$$ □

Question 14.1 A student designs an experiment to determine the specific latent heat of fusion L of ice. Some ice at 0°C is heated with an electric heater. The experiment is carried out twice and the following data are obtained.

	energy supply from heater (J)	time interval (min)	mass of ice melted (g)
heater off	0	10.0	14.3
heater on	21000	5.0	70.0

(a) Suggest why two sets of readings are taken. (b) Find specific latent heat of fusion for ice.

14.1.4 Specific heat capacity

heating a substance could cause an increase in its temperature

heat required is proportional to its mass m and temperature change ΔT: $\boxed{Q = cm\Delta T}$

> we define **specific heat capacity** (c) as the thermal energy required per unit mass of substance to cause an increase of one unit in its temperature

- unit of specific heat capacity: $[c] = \text{J kg}^{-1}\ \text{K}^{-1}$ or $\text{J kg}^{-1}\ {}^\circ\text{C}^{-1}$
- c is also an *intensive* property, i.e., independent of size or shape of the sample

Example 14.3 A block of 30 g ice at −20°C is added to a large cup of 270 g water at 80°C. Assume there is no energy lost, what is the final temperature of the mixture? (data: specific heat capacity of water is 4200 J kg^{-1} K^{-1}, specific heat capacity of ice is 2100 J kg^{-1} K^{-1}, specific latent heat of ice is 3.3×10^5 J kg^{-1}.

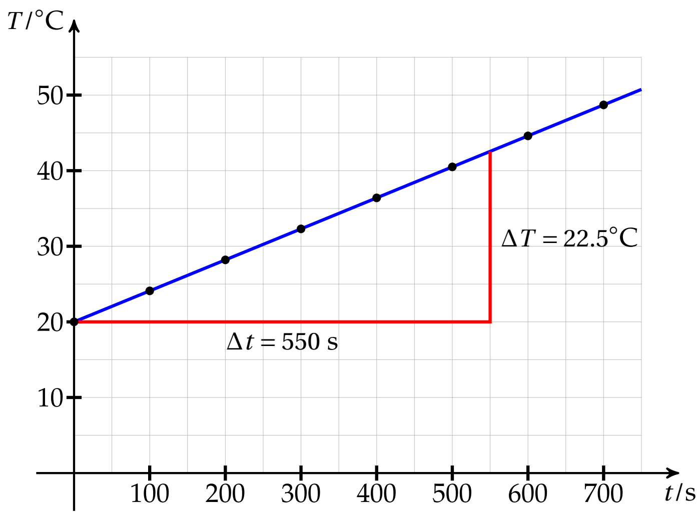
✍ energy lost by hot water = energy gain by ice cube

$$\underbrace{4200 \times 0.27 \times (80 - T)}_{95\ {}^\circ\text{C water} \to T\ {}^\circ\text{C water}} = \underbrace{2100 \times 0.030 \times [0 - (-20)]}_{-20\ {}^\circ\text{C ice} \to 0\ {}^\circ\text{C ice}} + \underbrace{3.3 \times 10^5 \times 0.030}_{0\ {}^\circ\text{C ice} \to 0\ {}^\circ\text{C water}} + \underbrace{4200 \times 0.030 \times (T - 0)}_{0\ {}^\circ\text{C water} \to T\ {}^\circ\text{C water}}$$

$$90720 - 1134T = 1260 + 9900 + 126T$$

$$T = \frac{90720 - 1260 - 9900}{1134 + 126} \approx 63°\text{C}$$ □

Example 14.4 A 1.00 kg aluminium block is heated using an electrical heater. The current in the heater is 4.2 A and the p.d. across is 12 V. Measurements of the rising temperature are represented by the graph. Determine specific heat capacity of aluminium.



✍ energy supplied: $Q = cm\Delta T \quad \Rightarrow \quad IV\Delta t = cm\Delta T \quad \Rightarrow \quad c = \dfrac{IV}{m\frac{\Delta T}{\Delta t}}$

$\dfrac{\Delta T}{\Delta t}$ is gradient of fitting line: $\dfrac{\Delta T}{\Delta t} = \dfrac{22.5}{550} \approx 4.09 \times 10^{-2}\ °\text{C s}^{-1}$

specific heat capacity: $c = \dfrac{4.2 \times 12}{1.00 \times 4.09 \times 10^{-2}} \approx 1230\ \text{J kg}^{-1}\ °\text{C}^{-1}$ □

Question 14.2 A mixture contains 5% silver and 95% of gold by weight. Some gold is melted and the correct weight of silver is added. The initial temperature of silver is 20°C. Use the data to calculate the initial temperature of gold so that the final mixture is at melting point of gold.

	silver	gold
melting point (K)	1240	1340
specific heat capacity (solid or liquid) (J kg^{-1} K^{-1})	235	129
specific latent heat of fusion (kJ kg^{-1})	105	628

14.2 Temperature & thermal energy

➢ temperature can be considered as a *relative* measure of thermal energy temperature can tell the *direction* of thermal energy flow

heat always (spontaneously) flows from high temperature regions to colder regions

➢ if two objects in contact have the same temperature, then there is no net heat transfer

the two objects are said to be in **thermal equilibrium**

➢ if two systems A and B are each in thermal equilibrium with a third system C, A and B are also in thermal equilibrium, this is called the **zeroth law of thermodynamics**

Question 14.3 A student thinks that temperature measures the amount of heat in an object. Suggest why this statement is incorrect with examples.

14.2.1 Absolute zero

mean K.E. of molecules is a microscopic description of temperature T

minimum K.E. occurs if molecules do not move at all (completely frozen)

this corresponds to the lowest possible temperature, called **absolute zero**

➢ **Kelvin scale of thermodynamic temperature** is defined based on absolute zero as 0 K

➢ conversion rule between Celsius scale and Kelvin scale: $\boxed{T_K(\text{K}) \underset{+273.15}{\overset{-273.15}{\rightleftharpoons}} T_C(^\circ\text{C})}$

➢ Kelvin scale is said to be an *absolute scale*

zero of Kelvin scale does not depend on property of a specific substance

in contrast, zero of Celsius scale is based on properties of water

➢ it is impossible to remove any more energy from a system at 0 K (or -273.15 °C)

but there is no practicable means to bring a physical system to exactly 0 K

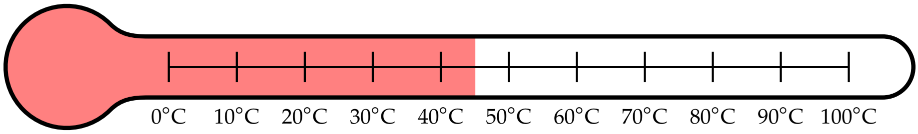
14.2.2 Thermometer

a **thermometer** is a device which can be used to measure temperature

Liquid-in-glass thermometer

basic principle: liquid expands in volume at higher temperature

examples include alcohol thermometer, mercury-in-glass thermometer, etc.



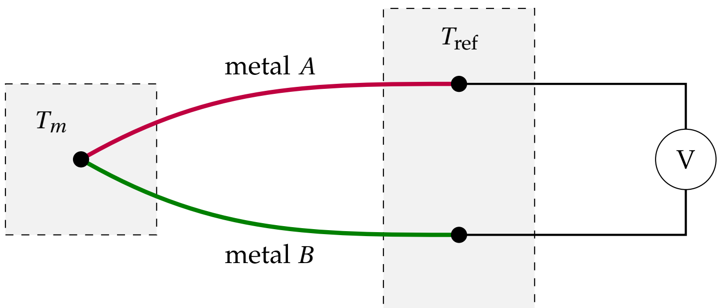
Resistance temperature detectors (RTD)

basic principle: resistance of electronic element changes with temperature

metal wires and thermistor are both used in RTD elements

Thermocouple

basic principle: difference in temperature can produce a *thermoelectric voltage* across junctions, thermocouple measures temperatures by means of this voltage



typical configuration of a thermocouple unit

two pieces of different metal wires are joined at their ends

if there exists a temperature difference between the ends, a *thermoelectric voltage* is developed

this voltage depends on temperature difference, captured by some characteristic function

in practice, we place the *measurement* junction in an environment of unknown temperature

the other end, or the *reference* junction, is at a known temperature

temperature difference is deduced from the voltage reading

hence the desired temperature can be determined

➢ features of a thermometer:

- *range*: whether the thermometer can measure very low or very high temperatures
- *sensitivity*: whether a small change in temperature can be detected
- *response time*: whether changes in temperature can be immediately measured
- *linearity*: whether changes in temperature are proportional to changes in output

	liquid-in-glass	RTD wire	thermistor	thermocouple
valid range	narrow	wide	narrow	very wide
sensitivity	low	fair	high	high
response time	slow	fast	fast	very fast
linearity	good	good	limited	non-linear